VOL. I · NO. 45 SUNDAY, 8 NOVEMBER 2026

The Rinfo

Research, in brief · the weekly review of new preprints from arXiv

WEEKLY EDITION · 01 NOV – 08 NOV 2026 · 64 PREPRINTS READ · 20 SELECTED

INSIDE THIS EDITION

CS.CL
Computation and Language
3 reports, 6 in brief

CS.LG
Machine Learning
3 reports, 6 in brief

EESS.SY
Systems and Control
2 reports, 0 in brief

SECTION · ARXIV CS.CL

Computation and Language

Coverage: conversational AI, dialogue systems, LLM agents and retrieval-augmented generation.

THE WEEK IN BRIEF

Recent research highlights significant advancements and security challenges in large language model agents and dialogue systems, focusing on proactive capabilities and memory frameworks such as Pro-Memory and MedGraph. Studies investigate multi-turn reinforcement learning, interactive backdoor vulnerabilities, and evaluation methods like tau-bench-LILT. Additionally, the literature addresses optimization techniques including pruning via selective knowledge informed masking and retrieval efficiency for conversational agents.

LEAD STORY · CS.CL · COMPUTATION AND LANGUAGE

Prospective Memory for Proactive LLM Agents

Researchers introduce Pro-Memory, a new framework that gives proactive large language model agents prospective memory to retain future intentions and decide when to act without explicit user requests.

BY YUANCHEN BEI, ZHINING LIU, TIANXIN WEI AND OTHERS · SUBMITTED 3 NOVEMBER 2026 · ARXIV:2611.05731

THE PROBLEM

Most existing memory systems for large language model agents are retrospective, retrieving past information rather than determining whether a future-directed intention remains pending and when the current situation warrants an action.

THE APPROACH. The authors propose Pro-Memory, a cognitively inspired framework that separates the maintenance of future intentions from deciding when and how to realize them by assessing new observations against original goals and activation conditions, updating intention states, and utilizing an experience memory for feedback.

KEY FINDINGS

- Pro-Memory separates maintaining future intentions from deciding when and how to realize them.
- The framework preserves pending objectives linked to supporting evidence and assesses new observations against original goals and activation conditions.
- Experiments across two benchmarks—spanning delayed-intention tasks and open-ended proactive assistance—demonstrate consistent improvements in proactive task performance over representative memory approaches.
- The approach limits unnecessary interventions across multiple large language model backbones.

WHY IT MATTERS

"For conversational AI, dialogue systems, LLM agents, and retrieval-augmented generation, this shifts agents from passive retrospective recall to active prospective realization, allowing them to provide timely assistance without explicit prompts while controlling disruptive interventions."

LIMITATIONS. The study focuses primarily on text-based proactive interactions, while real-world agents may require multimodal intention formation and cross-modal cue recognition, alongside complex long-horizon scenarios involving interdependent intentions and evolving user needs.

Filed under: proactive agents · prospective memory · large language models

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE GitHub - TeleAI-UAGI/Awesome-Agent-Memory: Curated systems, (github.com) · GitHub - TsinghuaC3I/Awesome-Memory-for-Agents: A Collection (github.com) · proactive-agents · GitHub Topics · GitHub (github.com)

REPORT · CS.CL · COMPUTATION AND LANGUAGE

Interactive Backdoor Attacks in LLM-Based Proactive Dialogue Systems

Researchers demonstrate that externally supplied dialogue data can be poisoned to implant interactive backdoors in large language model-based proactive dialogue systems, steering multi-turn conversations toward unauthorized attacker objectives.

BY ZHIYANG XIA, YUMING SHANG AND XI ZHANG · SUBMITTED 4 NOVEMBER 2026 · ARXIV:2611.06969

THE PROBLEM

Fine-tuning large language model-based proactive dialogue systems on externally supplied dialogue data introduces backdoor risks. Traditional backdoor studies focus primarily on immediate single-turn response deviations upon trigger activation, failing to account for multi-turn conversational guidance in proactive dialogue systems.

THE APPROACH. The authors present an interactive backdoor attack framework using training-data poisoning to implant a trigger-conditioned guidance-policy bias that ordinary user utterances can activate without deployment-time prompt modification. They instantiate three complementary attacks across progressively expanded interaction settings: Direct Attack, Constrained Progressive Attack, and Unconstrained Progressive Attack.

KEY FINDINGS

REPORT · CS.LG · MACHINE LEARNING

What Does Multi-Turn Reinforcement Learning Really Improve in LLM Agents?

New research investigates what multi-turn reinforcement learning actually improves in large language model agents, revealing that its performance gains depend heavily on sufficient interaction budgets rather than uniformly enhancing single-step decisions.

BY KEJIAN ZHU, ZHUORAN JIN, SHANGQING TU AND OTHERS · SUBMITTED 4 NOVEMBER 2026 · ARXIV:2611.06483

THE PROBLEM

The performance gains of multi-turn agentic reinforcement learning (RL) are highly sensitive to interaction constraints, and it remains poorly understood whether RL uniformly elevates single-step decision quality or specific capabilities like long-horizon resilience.

THE APPROACH. The authors evaluate agent performance across sampling breadth K and interaction depth T using $Pass@(K,T)$, analyze step-level action quality before and after multi-turn RL, provide a preliminary discussion on process supervision, and introduce BudgetMix RL to vary interaction budgets during training.

KEY FINDINGS

- Trajectory-level RL advantages depend on sufficient turn budgets and diminish significantly when budgets tighten.
- Agentic RL does not uniformly elevate single-step decision quality; its strength concentrates in error recovery and long-horizon resilience.

- All three proposed attacks effectively achieve their objectives under their respective interaction settings while largely preserving normal system utility.
- Experiments were conducted using two large language model backbones and two dialogue datasets within psychological counseling as a representative setting.
- Tested training-data filters retained nearly all poisoned dialogues.
- Three post-admission defense baselines left substantial residual attack-specific success.

LIMITATIONS. In the unconstrained progressive setting, silicon-sampled continuations remain an imperfect proxy that may share structural assumptions with the data-construction process, meaning the findings represent model-side steering within the sampled distribution rather than evidence of real-user susceptibility.

WHY IT MATTERS. It reveals a critical supply-chain security vulnerability in conversational AI and dialogue systems, demonstrating that data poisoning can manipulate sustained, multi-turn task guidance and conversational steering by LLM agents rather than just isolated single-turn outputs.

Filed under: backdoor attacks · dialogue systems · data poisoning

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE GitHub - NY1024/AgentSafety-Papers: Daily Tracking of LLM Ag (github.com) · Daily Papers - Hugging Face (huggingface.co) · GitHub - zuo-dot/Behavior-Safety-of-Autonomous-Interactive-A (github.com)

- RL-generated actions remain locally plausible under the base policy, but complete RL trajectories receive substantially higher likelihood under the RL policy.
- BudgetMix RL achieves robust performance across diverse execution settings by varying interaction budgets during training.

LIMITATIONS. The advantages of multi-turn RL diminish, disappear, or even reverse under tight interaction budgets, posing deployment risks in applications with strict constraints.

WHY IT MATTERS. Understanding how interaction budgets and multi-turn reinforcement learning affect LLM agents ensures that conversational and agentic systems are robust and reliable under realistic deployment constraints rather than evaluated solely on final task success.

Filed under: reinforcement learning · llm agents · interaction budgets

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE Paper page - What Does Multi-Turn Reinforcement Learning Rea (huggingface.co) · GitHub - WooooDyy/AgentGym-RL: Code and implementations for (github.com) · GitHub - yubol-bobo/Awesome-Multi-Turn-LLMs: This is the off (github.com)

IN BRIEF · MORE FROM CS.CL

MedGraph: Maintaining Patient State with Graph Memory for Long-Term Medical Dialogue

Donghua Zhang, Zizheng Wang and others · cs.CL · 5 November 2026

A Framework for Context Enrichment and Selective Robustness in LLM Social Agents

Mattia Greiche, Reihaneh Rabbany and others · cs.MA · 4 November 2026

SKIM: Pruning Large Language Model Agents via Selective Knowledge Informed Masking

Moonseok Choi, Jongwon Jeong and others · cs.CL · 5 November 2026

Learning When Human Decisions Matter: Executor-Relative Consultation Value for LLM Agents

Shuli Jiang, Khushbu Pahwa and others · cs.CL · 4 November 2026

Persistent Retrieval for Conversational Agents: A Study of Retrieval Efficiency

Karthik Ravula, Sathwik Kowtha and others · cs.AI · 5 November 2026

τ-bench-LILT: Evaluating Conversational Agents on Realistic Multilingual Tool-Use Tasks

Kaden Uhlig, Kiarash Mokhtari and others · cs.CL · 4 November 2026

ELSEWHERE IN THE JOURNALS

Peer-reviewed and web sources from the past month, beyond arXiv.

ACLANTHOLOGY.ORG

A Survey on Long-Term Memory Security in LLM Agents

Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Infor-mation Processing Systems, volume 33, pages 9459–9474. Curran Associates, Inc.…

ACLANTHOLOGY.ORG

Retrieval-Augmented LLM Agents: Learning to ...

models (LLMs) have ad-vanced the development of general-purpose agents, robust generalization to unseen tasks remains challenging. Two common approaches are supervised fine-tuning…

ACLANTHOLOGY.ORG

OP-Bench: Benchmarking Over-Personalization for ...

1 Introduction Recent advances in large language models (LLMs) have led to a new generation of memory-augmented conversational agents that can recall user prefer-ences, past…

HUGGINGFACE.CO

Daily Papers - Hugging Face

Retrieval-Augmented Generation (RAG) mitigates hallucination in large language models (LLMs) by incorporating external knowledge during generation. However, the effectiveness of…

SECTION · ARXIV CS.LG

Machine Learning

Also drawing on cs.AI, cs.CV, cs.CL, cs.DC. Coverage: energy-efficient and sustainable machine learning: lower compute, carbon and cost.

THE WEEK IN BRIEF

Recent research highlights a growing focus on energy-efficient and sustainable machine learning across various domains, ranging from large language model optimization to resource-constrained edge systems. Studies examine dynamic GPU energy reduction, medical question answering trade-offs, and the heterogeneous measurement landscape of TinyML. Furthermore, innovative edge deployment strategies and specialized hardware architectures continue to emerge to address compute, carbon, and cost constraints.

LEAD STORY · CS.DC · DISTRIBUTED, PARALLEL, AND CLUSTER COMPUTING

Energy-Efficient Reinforcement Learning for Large Language Models

Researchers from NVIDIA have introduced an online controller called EE-RL to reduce GPU energy consumption during reinforcement learning for large language models by dynamically adjusting clock frequencies.

BY RAHUL SHARMA, NITIN VEGESNA AND YANXIN ZHANG · SUBMITTED 3 NOVEMBER 2026 · ARXIV:2611.05748

THE PROBLEM

Reinforcement learning for large language models alternates between response generation and policy training, placing different demands on GPUs. Default GPU governors run all phases near peak clock frequency, causing memory-bound work and idle waiting to consume excess power without throughput gains.

THE APPROACH. The authors present EE-RL (energy-efficient RL), an online controller that selects separate clock frequencies for generation and training during execution. It compares candidate settings with the default GPU governor on the same token workload, keeps the most energy-efficient setting meeting a throughput threshold, and continuously adjusts frequencies as workloads change.

KEY FINDINGS

- Fixed-frequency sweeps on H100 and GB200 show GPU energy savings of up to 28.2 percent per valid training token across synchronous and asynchronous execution.
- EE-RL reduces GPU energy per valid training token by up to 16.9 percent at a modest throughput cost.
- Post-training Olmo 3 32B as a reasoning model used 17 times more datacenter energy than its instruction-tuned counterpart, driven largely by rollout generation.

WHY IT MATTERS

"The growing electricity demand of AI data centers makes energy efficiency crucial for operating costs and infrastructure expansion. By lowering GPU clock frequencies during phases with different arithmetic intensities, EE-RL reduces energy consumption and electricity costs without substantially extending GPU allocation time."

LIMITATIONS. Not stated in the provided text.

Filed under: reinforcement learning · gpu energy efficiency · dynamic voltage and frequency scaling

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE Daily Papers - Hugging Face (huggingface.co) · GitHub - yingchengyang/Reinforcement-Learning-Papers: Relate (github.com) · awesome-ai-efficiency/README.md at main · PrunaAI/awesome-ai (github.com)

REPORT · CS.CL · COMPUTATION AND LANGUAGE

Tug of Watts: A study on the Energy - Accuracy Trade-off of Large Language Models on Medical Question Answering Tasks

Researchers investigate the energy-accuracy trade-offs of large language models on medical question answering tasks, arguing that disjoint metrics are insufficient for sustainable pipeline design.

BY VIJU SUDHI, GIULIO BILLI, ANDREEA SCROB AND OTHERS · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.02889

THE PROBLEM

While large language models are increasingly deployed in medical question answering systems, studies often overlook their energy consumption and associated carbon footprints in favor of incremental improvements in downstream task accuracy.

REPORT · CS.LG · MACHINE LEARNING

Energy Measurement and Optimization of TinyML in Resource-Constrained Embedded Systems: A Systematic Mapping Study

A new systematic mapping study examines energy-aware TinyML research, highlighting a heterogeneous landscape lacking standardization in measurement and reporting.

BY CLARIELE ALMEIDA, RENAN BESERRA, RAFAEL MOURA AND OTHERS · SUBMITTED 5 NOVEMBER 2026 · ARXIV:2611.07264

THE PROBLEM

The literature on TinyML remains heterogeneous regarding evaluated algorithms, energy-measurement methods, reported metrics, influencing factors, and optimization strategies, alongside limited standardization that hinders reproducibility and direct comparison.

THE APPROACH. The authors examine 13 large language models across 3 model families ranging from 270M to 14B parameters on medical question answering tasks from the OpenMedical-LLM leaderboard. They use the CodeCarbon package to measure inference energy consumption and carbon emissions, and advocate for joint trade-off metrics and Pareto-optimal frontiers.

KEY FINDINGS

- Larger models attain higher task performance at the cost of higher relative carbon footprints.
- Reporting disjoint energy and accuracy metrics alone is insufficient for careful trade-off assessments.
- Systematic construction and interpretation of joint trade-off metrics are required for model evaluation.
- A distance-based rating scheme can help clearly distinguish the trade-offs of individual models.

LIMITATIONS. Not stated in the provided text.

WHY IT MATTERS. The study addresses the rising inference demands of AI systems—projected to reach 652 TWh by 2030—by helping practitioners treat model selection as a sustainability decision and jointly optimize energy consumption and accuracy.

Filed under: energy-accuracy trade-off · large language models · medical qa

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE Paper page - 70B-parameter large language models in Japanese (huggingface.co) · Paper page - PeFoMed: Parameter Efficient Fine-tuning on Mul (huggingface.co) · Paper page - Enhancing Healthcare through Large Language Mod (huggingface.co)

THE APPROACH. The authors present a systematic mapping of energy-aware TinyML research covering studies published between 2020 and 2025, identifying models and learning approaches, classifying energy-evaluation methodologies and metrics, examining technical factors affecting energy consumption, and organizing optimization techniques into a structured taxonomy.

KEY FINDINGS

- Frequent use of convolutional and lightweight architectures in energy-aware TinyML deployments.
- Reliance on direct measurements and analytical estimations to evaluate energy consumption.
- Identification of technical factors related to model architecture, hardware, communication, numerical precision, and input characteristics.
- Grouping of optimization strategies into model compression, hardware-aware design, system and communication optimization, and data and processing optimization.

LIMITATIONS. Limited standardization in experimental platforms, measurement procedures, and reporting practices, which constrains reproducibility and direct comparison across studies.

WHY IT MATTERS. Understanding and organizing energy-measurement methodologies, influencing technical factors, and optimization strategies helps reduce compute overhead, operational costs, and energy usage for resource-constrained edge devices.

Filed under: tinyml · energy efficiency · systematic mapping

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE GitHub - AI-in-Transportation-Lab/awesome-tinyml: A carefull (github.com) · TinyML_Papers/20250218_TinyML_Papers.md at main · OmniXRI/Ti (github.com) · omkar-jadhav-embedded-systems (Omkar Jadhav) · GitHub (github.com)

IN BRIEF · MORE FROM CS.LG

MedPsy: State-of-the-Art Small Medical Language Models for Efficient Edge Deployment

Davide Vitabile, Alexandro Buffa and others · cs.CL · 1 November 2026

uTECH Turbo-MOVES: The Intersection of Software Architectural Innovation and Sustainable Mobility Analysis

Cafer Avcı, Alireza Yazdiani and others · cs.SE · 2 November 2026

HyVox: Rethinking Low-Power Voice Intelligence towards Neuromorphic-von Neumann Edge Computing

Snehasis Addy, Eduardo Calle Ortiz and others · cs.NE · 1 November 2026

Anomaly Detection on Resource-Constrained Devices with Reservoir Computing

Marco Angioli, Mathias Persson and others · cs.LG · 3 November 2026

Superscalar Loop Processors

Luca Colagrande, Pascal Etterli and others · cs.AR · 1 November 2026

A Cross-Domain Matrix Engine for Signal Processing and Deep Neural Net Inference

Mehdi Safarpour · cs.AR · 2 November 2026

ELSEWHERE IN THE JOURNALS

Peer-reviewed and web sources from the past month, beyond arXiv.

SCIENCEDIRECT.COM

Toward sustainable on-device intelligence: A survey on energy-efficient RAG systems with small language models

Pillar 3: Green AI and Sustainability. A growing body of research has documented the environmental cost of the AI scaling paradigm. Strubell et al. showed that training a single…

MDPI.COM

An Explainable AI-Powered Carbon Prediction and Machine Learning for Carbon-Aware Tourism in the Community

36. Marmouzi, O.; Oumaira, I.; Ajana El Khaddar, M. A Systematic review of green and sustainable AI: Taxonomy, metrics, challenges, and open research directions.…

SCIENCEDIRECT.COM

AI's impact on energy efficiency and electricity consumption: Evidence from AI green computing power in China's National Green Data Centers

MDPI.COM

Artificial Intelligence Technology Embeddedness, Digital Transformation, and Sustainable Green Development | Sustainability | MDPI

This study pioneers the analysis of the impact of AI green computing power on urban electricity utilization efficiency and consumption. Based on 4519 observations from 279…

However, AI also introduces a new “carbon efficiency dilemma.” Rising energy consumption in computing infrastructure and data centers, the carbon footprint of model training,…

SECTION · ARXIV EESS.SY

Systems and Control

Also drawing on eess.SP, cs.LG, stat.ML, physics.app-ph, cond-mat.mtrl-sci. Coverage: battery management systems: state-of-charge and state-of-health estimation, degradation modelling and remaining useful life prediction.

THE WEEK IN BRIEF

Recent research in battery management systems and industrial time series highlights advances in machine learning applications. Investigators have introduced FlowBatt, a conditional generative framework utilizing flow matching and a diffusion transformer backbone for probabilistic battery degradation trajectory modelling. Additionally, studies on cross-domain self-supervised pretraining via JEPA-HITS demonstrate how shared encoder designs improve downstream regression and classification tasks when labeled data are scarce.

LEAD STORY · CS.LG · MACHINE LEARNING

FlowBatt: Flow Matching for Probabilistic Battery Degradation Prediction

Researchers have introduced FlowBatt, a conditional generative framework using flow matching and a diffusion transformer backbone to model probabilistic battery degradation trajectories.

BY HAMIDREZA EIVAZI, ESLAM SHARAAWY, STEFAN WITTEK AND OTHERS · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.03150

THE PROBLEM

Battery degradation prediction is difficult due to complex interactions among cycling behavior, operating conditions, and cell chemistry, while existing machine learning approaches largely focus on deterministic end-of-life predictions or fixed chemistry settings rather than probabilistic state of health trajectory modeling.

THE APPROACH. The authors introduce FlowBatt, a use-inspired conditional generative framework that adapts flow matching with a diffusion transformer (DiT) backbone to model full state of health (SOH) trajectories as probabilistic degradation processes conditioned on early-cycle capacity data. They also use explainable AI to diagnose and improve the encoder that maps early-cycle capacity matrices into conditioning vectors.

KEY FINDINGS

- FlowBatt achieves the best trajectory-wise SOH prediction performance on all five public benchmark datasets, with errors below 2.5%.
- The model delivers competitive remaining useful life (RUL) prediction performance compared to established baselines.
- By generating multiple plausible trajectories, FlowBatt provides empirical prediction intervals for battery degradation modeling.
- FlowBatt-generated SOH trajectories improve downstream RUL modeling, reducing random-forest RMSE and MAPE on MATR-1 by approximately 26% and 19%, respectively.

WHY IT MATTERS

“It advances battery management systems by shifting away from deterministic point estimations toward probabilistic, full-trajectory SOH modeling and uncertainty estimation across heterogeneous chemistries.”

LIMITATIONS. Prediction remains challenging on MATR-2 and HUST, resulting in higher RUL errors and more pronounced undercoverage of empirical prediction intervals. The model conditions only on capacity matrices from fixed 20- or 100-cycle windows without direct temperature, current-profile, charging-protocol, environmental, or cooling inputs, and fixed splits do not establish generalization to unseen chemistries or protocols.

Filed under: flow matching · battery degradation · state of health

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE 18839_Generative_Modeling_of_B.pdf (openreview.net) · GitHub - AkashDas-AI/Battery-Degradation-Prediction-for-Ener (github.com) · GitHub - HamidrezaEiv/DiffBatt: DiffBatt: A Diffusion Model (github.com)

REPORT · CS.LG · MACHINE LEARNING

Cross-Domain Joint-Embedding Pretraining For Heterogeneous Industrial Time Series

Researchers have investigated a cross-domain self-supervised pretraining approach called JEPA-HITS to learn reusable time-series representations across physically distinct industrial domains. The shared encoder design improves downstream regression and classification tasks when labeled data are scarce.

BY JONATHAN ADAM RICO, NAGARAJAN RAGHAVAN AND SENTHILNATH JAYAVELU · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.02489

THE PROBLEM

Industrial predictive maintenance and manufacturing tasks frequently contain abundant sensor measurements, but labeled data for degradation and faults are scarce and expensive. Transfer learning across physically distinct industrial domains remains underexplored because existing models often assume fixed input lengths, specific preprocessing, or rely on closely related datasets.

THE APPROACH. The authors investigate cross-domain self-supervised pretraining using a joint-embedding predictive architecture for heterogeneous industrial time series (JEPA-HITS). This framework learns a single encoder from diverse domains—such as battery, semiconductor chemical mechanical planarization (CMP), computer numerical control (CNC) machining, and bearing—while preserving dataset-specific temporal scales via multi-domain loss aggregation and a channel-independent latent encoder.

KEY FINDINGS

- Leave-one-domain-out (LODO) pretraining reduces root mean square error (RMSE) on battery remaining useful life (RUL) regression from 10.72 to 7.95 compared to a randomly initialized architecture.
- LODO pretraining reduces RMSE from 4.34 to 3.89 on semiconductor chemical mechanical planarization (CMP) and from 13.53 to 11.72 on tool wear regression.
- LODO pretraining improves bearing fault diagnosis macro-F1 score from 0.547 to 0.737.
- Unlabeled time series measurements from heterogeneous industrial processes provide reusable representations for downstream learning under reduced-label conditions.

LIMITATIONS. The work is not evaluated on extreme cases involving signal outliers, missing data, and distribution shifts.

WHY IT MATTERS. For battery management systems, the ability to leverage unlabeled multi-domain industrial time series data via a shared encoder offers a practical pathway to improve remaining useful life (RUL) and degradation modelling even when battery-specific labeled data are severely limited.

Filed under: battery rul · time series · self-supervised learning

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE Daily Papers - Hugging Face (huggingface.co) · Learning to Factorize Spatio-Temporal Foundation Models - NI (openreview.net) · GitHub - TSCenter/awesome-time-series-papers: An Awesome Lis (github.com)

ELSEWHERE IN THE JOURNALS

Peer-reviewed and web sources from the past month, beyond arXiv.

MDPI.COM

Fast State-of-Health Estimation of Lithium-Ion Batteries Using Short-Term Relaxation-Voltage Features Integrated with a Differential Evolution-Optimized Extreme Learning Machine Algorithm

27. She, C.; Li, Y.; Zou, C.; Wik, T.; Wang, Z.; Sun, F. Offline and Online Blended Machine Learning for Lithium-Ion Battery Health State Estimation. IEEE Trans. Transp.…

MDPI.COM

Capacity and SOH Prediction of Lithium-Ion Batteries Using Various Mathematical Forms of EIS with Neural Networks

Aloisio et al. showcased the utility of rectangular vs. polar coordinates for state of charge estimation . They argued that the phase angle of polar coordinates is more correlated…

SCIENCEDIRECT.COM

Electrochemical-guided time-frequency transformer for battery state-of-health estimation

Existing battery SOH estimation methods can be classified into physics-based and data-driven methods. The former model internal physical mechanisms to capture electrochemical…

SCIENCEDIRECT.COM

State of charge estimation in lithium-ion batteries: A critical ...

by EHE Bayoumi · 2026 · Cited by 3 — CCM is a widely used method for estimating the SoC in Li-ion batteries. The method operates by measuring the current flowing in or out of the…

COLOPHON

The Rinfo is compiled automatically each Sunday from the arXiv preprint server. Reports are written by a language model from each paper's own text and have not been reviewed by a human editor; they may contain errors. Preprints are not peer-reviewed. Consult the original paper before citing any claim. Links under "Also online" and "Elsewhere in the journals" are supplied by web search and do not imply endorsement.

Read online or download the PDF · Subscribe · Unsubscribe or change